P4: Two-Body Problem

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1 Instructions

Consider two masses, m_1 and m_2, under mutual gravitational influence. The position vectors for the two masses are $\mathbf{r}_1$ and $\mathbf{r}_2$. From Newton's second law and the universal law of gravitation, we get:

$$m_1\mathbf{r}''_1 = \mathbf{F}, \quad m_2\mathbf{r}''_2 = -\mathbf{F} \tag{1}$$

where

$$\mathbf{F} = -Gm_1m_2\frac{(\mathbf{r}_1 - \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3} \tag{2}$$

where G is the graviational constant. $\mathbf{r}''_1$ and $\mathbf{r}''_2$ are the second derivatives of $\mathbf{r}_1$ and $\mathbf{r}_2$ with respect to time.

The two-body problem can be reduced to a one-body problem by a change of coordinates. Let r be the relative position vector, and R be the coordinate of the center of mass of the system; that is:

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2, \quad \text{and} \quad \mathbf{R} = \frac{(m_1\mathbf{r}_1 + m_2\mathbf{r}_2)}{m_1 + m_2} \tag{3}$$

Now, changing coordinates in Newton's equations from $\mathbf{r}_1$ and $\mathbf{r}_2$ to $\mathbf{r}$ and $\mathbf{R}$ results in

$$\frac{m_1\mathbf{R} + (m_1m_2\mathbf{r})}{m_1 + m_2} = \mathbf{F}, \quad \frac{m_2\mathbf{R} - (m_1m_2\mathbf{r})}{m_1 + m_2} = -\mathbf{F} \tag{4}$$

and adding and subtracting these two equations yields the result

$$\mathbf{R}'' = 0 \quad \text{and} \quad \mu\mathbf{r}'' = \mathbf{F} \tag{5}$$

where μ is called the reduced mass and is given by $\mu = \frac{m_1m_2}{m_1+m_2}$.

The center-of-mass coordinate system is an inertial coordinate system (one moving at constant velocity with the center-of-mass fixed at the origin $\mathbf{R} = 0$). The remaining equation for the relative coordinate has the form of a one-body problem and can be written as

$$\mathbf{r}'' = -\frac{k\mathbf{r}}{r^3}, \quad \text{where} \quad k = G(m_1 + m_2) \tag{6}$$

Once $\mathbf{r}$ is determined, the coordinates of the two masses in the center-of-mass coordinate system can be found from

$$\mathbf{r}_1 = \frac{m_2 \mathbf{r}}{m_1 + m_2} \quad \text{and} \quad \mathbf{r}_2 = \frac{m_1 \mathbf{r}}{m_1 + m_2} \tag{7}$$

The equivalent one-body problem can be simplified further. The position vector $\mathbf{r}$ and the velocity vector $\mathbf{r}'$ form a plane. Since the acceleration vector points in the direction of the position vector, the one-body motion will be restricted to this plane. There is no force component trying to move the masses off the plane. We can place our coordinate system so that $\mathbf{r}$ lies in the x-y plane with $z = 0$ and write $\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$.

The relevant units are length and time, and they may be nondimensionalized using the constant parameter k (which has units l^3/t^2) and the distance of the closest approach of the two masses (the minimum value of $\mathbf{r}$). The dimensionless governing equations, in component form, are then given by

$$x'' = -\frac{x}{(x^2 + y^2)^{3/2}} \quad \text{and} \quad y'' = -\frac{y}{(x^2 + y^2)^{3/2}} \tag{8}$$

We can further orient the x-y axes so that the solution $r = r(t)$ is symmetric about the x-axis, and the minimum value of $\mathbf{r}$ occurs at $x = -1$ and $y = 0$, where symmetry also requires $x' = 0$. The quantities x' and y' are the first derivatives with respect to time. Setting our initial conditions at the point of minimum $\mathbf{r}$, we can write

$$x(0) = -1, \quad y(0) = 0, \quad x'(0) = 0, \quad y'(0) = (1 + e)^{1/2} \tag{9}$$

where we have parameterized $y'(0)$ using ε. The parameter ε is called the eccentricity of the orbit. Circular orbits have $\varepsilon = 0$, and elliptical orbits have $0 < \varepsilon < 1$. When $\varepsilon = 1$, the orbit is parabolic; for $\varepsilon > 1$, we get a hyperbolic orbit. The period of a closed orbit is given by

$$T = \frac{2}{(1 - e)^{3/2}} \tag{10}$$

NOTE: double check numerator of the fraction with prof/canvas

For this project, you will write code that solves the two-body problem. You will investigate the orbit for four mass ratios (m_1:m_2) – 1:1, 1:2, 1:4, and 1:16. For each mass ratio, you will determine the orbit for the four eccentricity values – 0, 0.25, 0.50, and 0.75. Altogether, you will have 16 configurations.

You can write your own Runge-Kutta fourth-order solver or use the ODE solver in `SciPy`. Given the initial conditions, specified value of e, and mass ratio, solve a system of four first-order differential equations for the sixteen configurations.